CSE 318 — Online 2

Section A2

In your Greedy heuristic, after placing the two endpoints of the heaviest edge, the remaining vertices are considered in some fixed order (e.g., by index). The *order* in which vertices are processed can significantly affect the final cut.

Your task: Instead of a fixed order, at each step, **dynamically pick the vertex that has the most to lose from a wrong placement**. We call this the "regret" of a vertex:

1. Start exactly as your greedy: place the endpoints of the heaviest edge into X and Y.
2. At each step, for every unassigned vertex v, compute:
   - $\sigma_X(v)$ = total weight of edges from v to vertices already in X
   - $\sigma_Y(v)$ = total weight of edges from v to vertices already in Y
   - $\text{regret}(v) = |\sigma_X(v) - \sigma_Y(v)|$

   The regret measures how much it matters to get this vertex right. If $\sigma_X(v)$ and $\sigma_Y(v)$ are close, it does not matter much which side v goes to. If they are very different, it matters a lot.
3. Pick the unassigned vertex with the **largest regret**. Break ties by choosing the vertex with the larger $\max\{\sigma_X(v), \sigma_Y(v)\}$, and further ties by smaller vertex index.
4. Place it using the greedy rule: put v in X if $\sigma_Y(v) > \sigma_X(v)$, otherwise put v in Y.
5. Repeat until all vertices are placed.

What to run and report:

- Run the regret-based greedy on at least 6 benchmark graphs.
- Run your original greedy (fixed vertex order) on the same graphs.
- Report the cut value from the construction (before local search) and also after local search for both.